

Clinical Risk Assessment Report: Patient 26

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.52x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 4.0%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.5 [Bottom 10%] (+1.3%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.29 [Avg (28th %ile)] (+0.9%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.63 [Bottom 24%] (-4.0%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.46 [Avg (67th %ile)] (-1.2%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.2 [Avg (46th %ile)] (-1.0%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.44 [Avg (39th %ile)] (-0.9%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

PET Radiomic Metabolic Feature (GLZLM_ZLNU.PET) **Value: -0.51 [Bottom <1%] (-0.6%)**
Metabolic PET feature reflecting glucose avidity and cellular metabolic rate.

Patient Age (Age) **Value: 59.0 [Avg (33th %ile)] (-0.6%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Machine Learning Feature Attribution (SHAP Decision Path)

